BACHELOR OF TECHNOLOGY IN ARTIFICIAL
INTELLIGENCE

PURBANHAL UNIVERSITY
(FACULTY OF SOENCE AND TECHNOLOGY)
BIRATNAGAR, NEPAL

REGULATIONS GOVERNING BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE PROGRAM

1. TITLE OF PROGRAM:

The programme shall be called BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE.

2. ELIGIBILITY FOR ADMISSION:

Students seeking admission in BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE program:

2.1 Should have successfully completed twelve years of schooling in science stream.

2.2 Should have achieved minimum D+ grade in each subject of 10+2 with CGPA 2.0 or more

or

Should have secured a minimum score of second division (45%) marks in 10+2, PCL or equivalent in science discipline.

Students who have passed grade 11 and are waiting for supplementary exam (PURAK PARIKSHA) of grade 12 can also apply. However, they have to submit all the required documents at the time of admission.

Students who appeared in the final exam and are waiting for the result and certificates can also apply for the entrance examination. However, they have to submit all the required documents at the time of admission.

2.3 In case of foreign certificate, student should submit equivalence certificate and each subject grading with CGPA or total percentage document from concerned authority.

2.4 Should pass entrance examination as conducted by Purbanchal University.

3. DURATION OF THE PROGRAM:

The program of study shall extend over a period of eight semesters (FOUR ACADEMIC YEARS).

4. MEDIUM:

ENGLISH shall be the medium of instruction and examination in all the subjects of the Program.

5. **ATTENDANCE REQUIREMENT**:

A student must achieve, at least 80% attendance of lectures, tests, and tutorial classes in order to qualify for sitting for the final examination of any subject.

There are no unauthorized cuts from classes; persistent poor attendance may result in exclusion from classes.

In the case of unavoidable absence such as for illness of the student or serious illness or death of a member of the family or similar compelling reasons for absence, all works missed must be satisfactorily made up and the responsibility for making up this work rests with the concerned students.

Teachers should also help them in making up this work.

6. EVALUATION PROCEDURES:

(a) CONTINUOUS ASSESSMENT

All courses undertaken by students are evaluated during semester using an internal system of continuous assessment.

The student is evaluated on class and/or tutorial participation, assignment work, laboratory work, class tests and quizzes that contribute to the final grade awarded for the subject.

Students will be notified at the commencement of each course about the evaluation methods to be used for the course and the weightage given to the different assignments and evaluation activities.

(b) COMBINED THEORYAND LABORATORY/PRACTICAL COURSES:

Some of the courses have combined theory and laboratory/practical portions. For these courses marks will be awarded as follows: 20% Internal Marks, 20% Practical Marks, 60% Final Examination Marks.

The type of each course is indicated in the following course descriptions.

(c) THEORY:

The pure theory course marks will be awarded as follows:

20% Internal Marks and 80% Final Examination Marks.

(d) LABORATORY:

The pure laboratory or practical course marks will be awarded as follows:

60% from continuous internal evaluation, 40% from final viva to be evaluated by the University.

(e) END-SEMESTER EXAMINATION:

The examination at the end of the semester is set and evaluated by examiners.

7. OBJECTIVE OF SEMESTER COMPUTER PROJECTS:

The concepts of project work will begin in the second semester and it will continue in the last Seven semesters. Students will be expected to apply the theory and principles they have learned from other courses in a practical way in order to complete a project each semester. They will develop skills in goal setting, planning, research, team work, implementation, assessment, report writing and presentation is they work on their chosen project.

Student will work in a group of upto three students under the guidance of group advisor. The group will decide on a project and set out their aims and objectives.

8. EVALUATIONS AND GRADING SYSTEM:

The performance of students is evaluated through a system of continuous testing spread over the entire period of study. At the end of each semester, students are awarded letter grades based on grades and marks obtained in various segments of the course evaluation.

In students rating eight grades A+,A,B+, B, C, D ,F and I are used.
Letter grades are used to show the academic standing of a student, with the following values, equivalent marks % and remarks:

EQUIVALENT MARKS %	LETTER GRADE	GRADE VALUE	REMARKS
90 and Above	A+	4.00	
80 and Below 90	A	3.75	
70 and Below 80	B+	3.50	
60 and Below 70	B	3.00	
50 and Below 60	C	2.50	
40 and Below 50	D	1.75	
Below 40	F	0.00	Fail
Not Qualified(NQ)/Absent	I	-	Incomplete

If a student fails to submit term paper, report, home assignment and laboratory assignment, which are requirements of a course, the teacher concerned may allow him the benefit of an "Incomplete." A student who is awarded as "Incomplete" in any course can get it removed within six weeks from the end of the semester. If the requirements are not met within this time limit, the student's grade in that course is converted into "Fail." On completion of the course, however, the student does not receive any further grade but is allowed the benefit of the numerical grade point weight of an "Incomplete."

9. **CUMULATIVE GRADE POINT AVERAGE (CGPA)**

A Cumulative Grade Point Average (CGPA) , Which is the grade point average of all the semesters, is computed at the end of the course for all students. Final later grades in each course are converted into grade points on the following basis:

A+--- 4.00 grade points
A- 3.75 grade points
B+--3.50 grade points
B-- 3.00 grade points
C---2.50 grade points
D--1.75 grade points
F---0.00 grade points

As the student complete different course, these points are accumulated and an average point score for each student called the CGPA in maintained.

The CGPA shall be calculated using the for formula:

$$CGPA = \frac{\Sigma(\text{Credit hours} \times \text{Grade points})}{\Sigma(\text{Credit hours})}$$

A student must maintain a CGPA of 2.0 or above throughout the study period. The student failing to maintain the CGPA of 2.0 may be required to withdraw from the program.

10. SCOPE FOR FURTHER STUDIES:

After accomplishing this program, the student can enroll for graduate degree such as:

Master of Technology in Artificial Intelligence
Master of Science in Artificial Intelligence
Master of Computer Application
Maser of Information Technology
Master of Science in Computer Science
Master of Science in Computer Information Systems
Master of Business Administration

11. THE DISTRIBUTION OF COURSE SHALL BE AS FOLLOWS:

Year:I **Semester:I**

Course Code	CourseTitle	Credits	Lecture (Hrs.)	Tutorial (Hrs.)	Laboratory (Hrs.)	Total (Hrs.)
BT101CO	Introduction to Programming	3	3	1	2	6
BT102CO	Digital Logic	3	3	1	2	6
BT103HS	Calculus	3	3	1		4
BT104HS	Introduction to Artificial Intelligence	3	3	1		4
BT105HS	Technical Report Writing and Presentation	3	3	1		4
BT106HS	Society and Professional Ethics	3	3	1		4
	Total	18	18	6	4	28

Year:I **Semester:II**

Course Code	Course Title	Credits	Lecture (Hrs.)	Tutorial (Hrs.)	Laboratory(Hrs.)	Total (Hrs.)
BT251CO	Object oriented Programming	3	3	1	2	6
BT252CO	Artificial Intelligence and Intelligent Systems	3	3	1	2	6
BT253CO	Microprocessor and Assembly Language	3	3	1	2	6
BT254CO	Probability and Statistics	3	3	1		4
BT255HS	Linear Algebra	3	3	1		4
BT256CO	Project I	2	1		2	3
	Total	17	16	5	8	29

Year:II **Semester:I**

Course Code	Course Title	Credits	Lecture (Hrs.)	Tutorial (Hrs.)	Laboratory (Hrs.)	Total (Hrs.)
BT201CO	Database Management System	3	3	1	2	6
BT202CO	Data Structure and Algorithm	3	3	1	2	6
BT203CO	Computer Organization	3	3	1		4
BT204CO	Operating system	3	3	1	2	6
BT205HS	Differential Equations	3	3	1		4
BT206CO	Project II	2			3	3
	Total	17	15	5	9	29

Year:II **Semester:II**

Course Code	Course Title	Credits	Lecture (Hrs.)	Tutorial (Hrs.)	Laboratory (Hrs.)	Total (Hrs.)
BT251CO	Numerical Methods	3	3	1	2	6
BT252CO	Computer Networks	3	3	1	2	6
BT253CO	Design and Analysis of Algorithms	3	3	1		4
BT254CO	Programming for AI (Tools, Techniques and Libraries)	3	3	1	2	6
BT255HS	Introduction to Data Science	3	3	1	2	6
BT256CO	Project III	2			3	3
	Total	17	15	5	11	31

Year III **Semester I**

Course Code	Course Title	Credits	Lecture (Hrs.)	Tutorial(Hrs.)	Laboratory(Hrs.)	Total (Hrs.)
BT301CO	Machine Learning	3	3	1	2	6
BT302CO	Optimization Techniques	3	3	1	2	6
BT303CO	Embedded Systems	3	3	1	2	6
BT304CO	Pattern Recognition and Image Processing	3	3	1	2	6
BT305CO	Data warehousing and Mining	3	3	1	2	6
BT306CO	Project IV	2			3	3
	Total	17	15	5	13	33

Year III **Semester II**

Course Code	CourseTitle	Credits	Lecture (Hrs.)	Tutorial(Hrs.)	Laboratory (Hrs.)	Total (Hrs.)
BT351CO	Object Oriented Software Engineering	3	3	1	2	6
BT352CO	Computer Vision	3	3	1	2	6
BT353CO	Deep Learning	3	3	1	2	6
BT354CO	Speech and Natural language Processing	3	3	1	2	6
BT355HS	Research Methods	3	3	1		4
BT356CO	Project V	2			3	3
	Total	17	15	5	11	33

Year IV **Semester I**

Course Code	Course Title	Credits	Lecture (Hrs.)	Tutorial (Hrs.)	Laboratory (Hrs.)	Total (Hrs.)
BT401CO	Internship	3			3(Practical and Field work)	3
BT402CO	Reinforcement Learning	3	3	1	2	6
BT403CO	Cloud Computing	3	3	1	2	6
BT404CO	Project VI	2			3	3
	Total	11	6	2	10	18

Year:IV **Semester:II**

Course Code	Course Title	Credits	Lecture (Hrs.)	Tutorial (Hrs.)	Laboratory (Hrs.)	Total (Hrs.)
BT451CO	IoT for Smart city	3	3	1	2	6
BT452**	Elective I	3	3	1		4
BT453**	Elective II	3	3	1		4
BT454CO	Professional Project	3		1	3	4
	Total	12	9	4	5	18

List of Elective Courses

Computer Vision	Image and Video Processing Surveillance Video Analytics People detection and Bio-metric Recognition
Mathematics	Introduction to Statistical Learning Optimization methods in Machine Learning/ Convex Optimization Kernel Methods Bayesian Data Analysis
Computing/ IoT	Smart Product Development Predictive Analysis and IoT Modeling & Simulation Pervasive Computing
Data Science	Fundamentals of Big Data Analytics Data Analysis and Visualization Business Intelligence Social media Analytics